

Voice & Speech Profile

Diagnostic summary, training plan, and app specification | Adarsh | August 2026

If you are reading this a year later and remember nothing: there was never anything wrong with your voice. Your vocal folds seal properly, your breath capacity is normal, your endurance is normal. What you had was a **habit** — you spoke at roughly twice the air pressure the job required, all day, every day. Every symptom you noticed (running out of breath, sounding harsh, merging words, being unable to speak quietly, sounding tired) was a downstream consequence of that single habit.

The proof is one number: speaking a sustained vowel at your normal volume lasted **15-16 seconds**. The same vowel at a soft volume lasted **25 seconds**. Same lungs, same day, 60% more output. Nothing anatomical changed in between. Only the pushing did.

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1. The complaints, as originally described

These were the symptoms noticed before any measurement. Recorded verbatim in substance so that future-you can check whether they still apply.

Complaint	What it felt like
Effortful speech	Speaking felt like work. Fatigue after long conversations or extended talking.
Running out of air	Breath ran short mid-sentence; sentences had to be cut or restarted.
Words merging	Consecutive words "eating" each other — endings of words disappearing into the next word.
Heavy / thick voice	Self-described as "moti awaaz" — a heavy, thick sound. Wanted something deeper, calmer, less effortful.
No quiet gear	In a silent room, speaking softly to one person is difficult — the voice either doesn't come out at all, or comes out uneven and breathy.
Harsh, not polite	Default tone reads as blunt or harsh. Anger projects easily and convincingly; softness and warmth do not. Emotional range in ordinary conversation feels narrow.
Only one emphasis dial	To stress an important word, the only available tool is getting louder.

2. The tests, the numbers, and what each one rules in or out

Five measurements were taken. Each one exists to eliminate a specific possible cause. Read this table as a process of elimination, not as a scorecard.

Test	Result	Normal range	What it rules in / out
Counting on one breath (one comfortable breath, count aloud at normal pace)	28	Adult male: 25–40	Breath support and lung capacity are normal . Rules out "small lungs" or a breathing-mechanics problem as the primary cause.
Sustained /s/ (voiceless hiss)	18 sec	15–25 sec	Airflow control without voicing is normal. This is the "engine only" baseline.
Sustained /z/ (voiced buzz)	25 sec	15–25 sec	Airflow control <i>with</i> the folds vibrating is normal — in fact better than the unvoiced one.
s/z ratio (18 ÷ 25)	0.72	Healthy < 1.4	The single most important number here. A ratio above 1.4 means air is leaking through folds that fail to close — the classic signature of nodules, polyps, paresis, or a glottal gap. At 0.72 you are at the opposite extreme. Your folds close efficiently. Closure pathology is ruled out.
Maximum Phonation Time — open vowel "aaah", habitual volume	15–16 sec	Adult male: 20–30 sec	Poor — and the only poor result in the entire set. This is the finding that needed explaining.
MPT — same vowel, deliberately soft volume	25 sec	—	Normal. The contradiction resolves here. A 60% improvement from changing exactly one variable: how hard you pushed.

The contradiction, and why it was the key

Three of the results were good and one was bad, all from the same set of lungs and the same pair of vocal folds on the same day. That is not how tissue damage behaves — damage is present in every task. So the question became: what is different about the one task that failed?

Answer: it was the only task with no back-pressure.

- **Counting** has consonants and micro-stops built into it. Every plosive briefly interrupts the airstream. The task self-regulates your output.
- **/z/** is a *semi-occluded* sound — the narrow constriction at the teeth creates pressure above the vocal folds, which props them into efficient, low-effort vibration. This is the exact mechanism behind straw phonation therapy. You accidentally performed a therapeutic exercise and scored well on it.
- **Open "aaah"** has nothing restraining the airflow. Nothing is holding you back — and the moment nothing holds you back, you dump air.

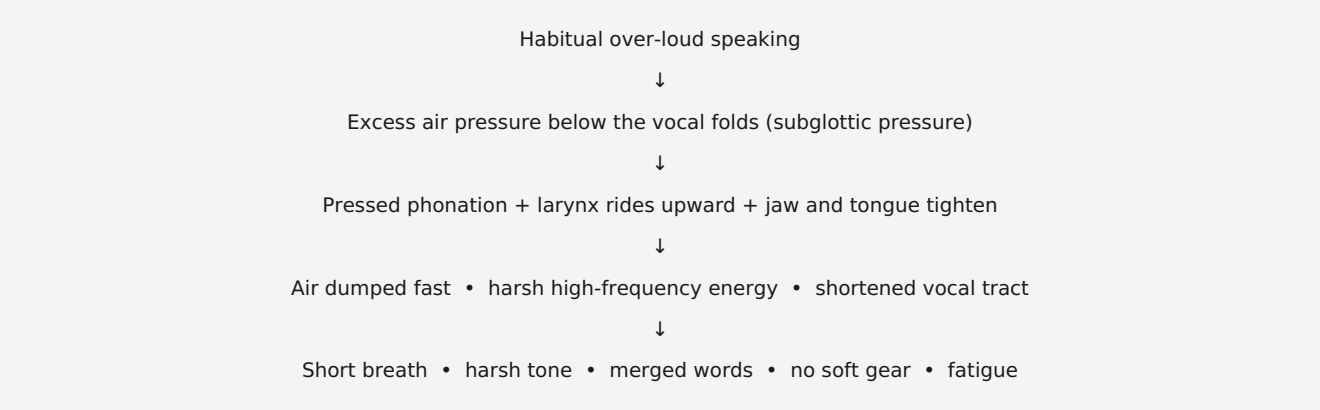
The pattern is unambiguous. When something restricts your airflow, you are efficient. When the road opens up, you floor it. That is not a capacity problem; it is a **drive** problem.

3. The diagnosis — one root cause, five downstream symptoms

ROOT CAUSE

Chronic excess subglottic pressure — a habitual over-drive of the voice. You speak at a volume and air pressure well above what the situation requires. Your habitual "comfortable" volume is other people's loud. This is a learned motor habit, not a condition, not a defect, and not damage.

The causal chain



How each symptom traces back to the root cause

Symptom	Mechanism
1. Running out of breath	You are spending air at roughly 1.6× the necessary rate. It isn't that the tank is small; the throttle is wide open. Confirmed directly: 15 sec loud vs 25 sec soft.
2. Words merging / consonants disappearing	This is acoustic masking . A loud, energy-dense sound does not stop the instant you stop producing it — its energy persists briefly and drowns out whatever comes immediately after. The quietest and shortest sounds in speech are precisely the word-final consonants: t, d, k, p, s. Those consonants <i>are</i> the boundaries between your words. A louder voice smears forward more, so it needs more space around each word , not less. You were giving it less, by speaking fast. This is why "slow down" is not generic advice in your case — it is compensation for a specific acoustic property of your own voice.
3. Harsh tone	Pressed phonation generates <i>more</i> high-frequency energy, not less. High-frequency energy is what the ear reads as harsh, sharp, and effortful. The harshness is a by-product of pressure, not of pitch or of personality.
4. Larynx elevation, and the loss of perceived depth	Under high drive, the larynx climbs. A raised larynx <i>shortens</i> the vocal tract, which raises resonant frequencies — the same principle as a flute versus a bassoon. So the pushing makes you sound less deep, while costing you effort. You have been paying for the opposite of what you wanted.
5. No quiet register	Covered in full in Section 6. In short: a skill you have never practised, because you have never needed it.
6. Only one emphasis dial	If loudness is your only expressive variable, then every emphasis must be a volume increase — which raises the average, which reinforces the habit. Pitch, pace, and pause are the missing dials.

4. What was ruled out, and why that matters

Write this down clearly, because in a year you may start worrying about it again from scratch.

Possible cause	Status	Evidence
Vocal fold closure problem — nodules, polyps, cyst, paresis, glottal gap	Ruled out	s/z ratio of 0.72, far below the 1.4 threshold. Folds close efficiently, possibly firmly.
Insufficient breath support or lung capacity	Ruled out	Count of 28 on one breath, within normal range.
Poor vocal endurance / fatigue-limited voice	Ruled out	/s/ 18 sec and /z/ 25 sec, both normal.
Anatomically limited pitch	Not a defect	Fundamental frequency is set by vocal fold length and mass. In an adult man it is essentially fixed. This is normal anatomy, not a problem to solve.
Habitual over-drive (excess subglottic pressure)	Confirmed	The 15 sec → 25 sec jump on a single changed variable. Plus the semi-occluded /z/ scoring better than the open vowel.
Untrained prosody and tonal control	Confirmed	Self-reported: anger projects, softness does not. No history of deliberate practice. See Section 7.

Note on the straw-in-water test. Its timing result is not comparable to open MPT — part of your air budget is spent maintaining bubble pressure in the water, so the seconds mean something different. Use the straw as an *exercise*, not as a measurement. And do not expect a single straw session to transfer to speech; transfer takes weeks of repetition.

5. The two perception traps

Trap 1 — bone conduction: your voice sounds deeper to you than it does to anyone else

When you speak, you hear yourself through two paths: air conduction (what everyone else hears) and bone conduction through your skull. Bone conduction adds low-frequency energy that no listener receives. Your voice therefore sounds thicker and bassier to you than to the room. When you record yourself, expect it to sound thinner and sharper than you remember — that recording is the true one, and the surprise is normal, not a sign of a problem.

Consequence: your subjective sense of "heavy" is partly an artefact of your own skull. Do not tune your voice by internal feel alone. Use recordings.

Trap 2 — "heavy" and "deep" are not the same thing

These are two independent dials, and conflating them was the core confusion in the original question.

	Pitch (fundamental frequency, F0)	Resonance
Determined by	Length and mass of the vocal folds	Shape of the throat, mouth, and pharynx
Changeable?	Barely — it is anatomy	Substantially — it is posture
Gives you	The note	The perceived richness, warmth, and depth
Your lever?	No	Yes — this is where all the gains are

The honest answer to "can I get a deeper voice":

- **Pitch: no**, not meaningfully. Anyone selling a pitch-lowering method is selling you vocal fry.
- **Perceived depth: yes, substantially.** Most of what a listener hears as "deep" is resonance, and resonance is posture — larynx height, pharyngeal space, and chest engagement are all trainable.

What you were feeling as heaviness was, most likely, **loudness plus tension** — not genuine low-end richness. Removing the tension will give you more actual depth than any amount of pushing did.

6. Sub-problem A — the missing quiet register

The complaint: in a completely silent room, speaking softly to one person is hard. The voice either doesn't come out, or comes out uneven, breathy, or patchy.

Why this happens

Soft speech is not simply "the same thing, turned down." It is mechanically a different task, requiring a different coordination:

- **Loud speech** is high air pressure plus firm fold closure. Momentum does the work. It is forgiving of sloppy coordination — pressure covers up the errors.
- **Soft speech** is low air pressure plus *precisely* the right amount of closure. Too little closure and the tone breaks into breathiness; too much and the sound refuses to start at all. There is no momentum to hide behind, so control has to be exact.

You have spent years training only the loud coordination. The quiet one has essentially never been practised, so the fine motor control for it does not yet exist. This is why softness feels like a different instrument — functionally, it is.

The good news: this is the most trainable thing in the whole document, because it is pure skill acquisition, starting from zero, with no anatomical obstacle in the way. Expect noticeable change within 3–4 weeks of daily practice.

The specific fix — the volume ladder

The goal is to build a continuous, controllable range instead of a single loud setting with a cliff below it. Practised daily in Block C of the routine (Section 8).

Level	Label	Use case and target feel
1	Near-whisper, but <i>voiced</i>	Silent room, person beside you. Must still have tone — a whisper is not level 1. This is the hardest level and the one you're missing.
2	Quiet conversational	Two people, quiet room, small table. Your realistic everyday target.
3	Normal conversational	Small group, ordinary room. Probably where your current level 2 sits.
4	Room-filling	Meeting, presentation, 10+ people. Probably where your current default sits.
5	Projection	Noisy environment, large room. Should be rare and deliberate.

The core insight to remember: your current everyday setting is roughly a 4. The goal is to make 2 your default and to make 1 available on demand. Levels 4 and 5 are not being taken away from you — they are being made into choices instead of the only option.

7. Sub-problem B — sounding harsh instead of polite

The complaint: the default delivery reads as harsh or blunt rather than polite. Anger projects easily and lands accurately. Softness, warmth, and gentleness do not. The suspicion — correct — is that this is a practice problem, not a voice-sound problem.

Two contributing layers

Layer 1 — acoustic (partly the over-drive). Pressed phonation adds high-frequency energy, and hard onsets (starting a vowel with a small glottal click) read as abrupt. This part improves automatically as the volume habit fades. It is maybe 40% of the problem.

Layer 2 — prosodic (untrained skill). The other 60%. "Politeness" in speech is not a quality of the voice itself; it is a specific, learnable pattern of pitch, pace, and pause. Nobody ever taught you the pattern, and you have never rehearsed it, so it simply is not in your motor repertoire. Anger *is* in your repertoire — because anger's pattern (loud, fast, hard onsets, flat-then-spiking pitch) is very close to your default. That is the whole explanation for the asymmetry.

What actually makes speech sound polite — the six parameters

Parameter	Harsh / blunt setting	Polite / warm setting
Onset	Hard glottal attack — the word starts with a click or punch	Soft onset — a breath of air begins fractionally before the tone. Practise by starting words with a barely-audible "h" and then removing the h while keeping the feel.
Terminal pitch	Flat or clipped endings; abrupt stops	Gentle falling contour on statements. Falling ends read as calm and settled. (Rising ends read as uncertain — a different problem, not yours.)
Pitch variability	Narrow range, which reads as flat, cold, or impatient	Moderate movement across the phrase. Warmth lives in melodic variation, not in volume.
Pace	Fast, especially at the start of a sentence	Slower onset into the sentence; the first three words set the perceived attitude for the whole utterance.
Pause	Emphasis achieved by getting louder	Emphasis achieved by pausing <i>before</i> the important word. This is the single highest-leverage swap available to you — it replaces your only emphasis dial with a better one and lowers your average volume at the same time.
Volume floor	Sits at level 4	Sits at level 2. Softness itself is read as consideration for the listener.

The emotional palette drill

You can already produce anger. The task is to build the other colours with the same reliability. Take one neutral sentence — for example, "*I'll look at it tomorrow morning.*" — and deliver it in each mode, recording all of them in one take so they can be compared back to back:

1. **Neutral / informational** — the baseline.
2. **Warm / friendly** — soft onsets, more pitch movement, slower start.
3. **Reassuring** — lower volume, slower, falling terminals, longer pauses.
4. **Firm but polite** — the hardest and most useful one. Steady pitch, deliberate pace, no volume increase at all. This is the register that most professional situations actually require.
5. **Apologetic / softening** — quietest, slowest, highest pitch variation.
6. **Enthusiastic** — fast and high-variation, but explicitly *without* raising volume. This one directly attacks the loudness-equals-energy reflex.

Play the six back. Early on, several will sound nearly identical to each other — that is the finding, not a failure. The measure of progress is that a listener could identify each one blind.

8. The daily routine (12 minutes)

Ordered deliberately. Release comes before production; the transfer moment at the end of each block is where the actual learning happens. **Straw work is step one, not step two** — the /z/ result already proved that semi-occluded work is effective on you specifically.

Block A — Release (2 minutes)

Exercise	Dose	How, and what "correct" feels like
Yawn-sigh	6 reps	Begin a real yawn. Feel the larynx drop and the throat open into a wide, cool space. Before the yawn completes, sigh out on "haaah" while holding that open feeling. <i>That dropped position is the target posture for everything else.</i>
Jaw release	30 sec	Let the jaw hang — it should feel loose and heavy, not held. Massage the masseter (the muscle bulge above the back molars). A clamped jaw at speed is the direct cause of consonant collapse.
Tongue forward	30 sec	Tongue tip resting behind the lower front teeth, body low and forward, not bunched back toward the throat. Bunching backward is a common tension pattern that muddies vowels.
Neck & larynx check	15 sec	Fingers lightly on the front of the throat. Speak a short sentence. If the larynx rides up, that is the habit showing itself. You are not correcting it here — only noticing it.

Block B — Semi-occluded work (SOVT) (3 minutes)

Exercise	Dose	How, and what "correct" feels like
Straw phonation	2 min	Hum a comfortable steady note through a narrow straw. Then glide slowly up and down through your range. Should feel almost effortless, with a buzz felt in the lips and face. If it feels like work, you are still pushing.
Straw into water	30 sec	Straw in about 2 inches of water. Hum so the bubbles stay <i>steady and even</i> — that evenness is the real objective, not the duration.
The transfer moment	30 sec	The most important 30 seconds of the routine. Remove the straw and immediately speak one ordinary sentence, trying to keep the exact same easy feeling. The straw is not the exercise — carrying its feeling into speech is the exercise. Nothing here matters if this step is skipped.

Block C — The volume ladder (3 minutes)

Exercise	Dose	How, and what "correct" feels like
Descend the ladder	5 reps	Same sentence at level 5, then 4, 3, 2, 1. Each step should be a distinct, deliberate change — not a smooth fade. Then reverse: 1 up to 5.
Level 1 hold	1 min	Speak continuously at level 1 — genuinely quiet, but <i>voiced</i> , never whispered. When it breaks up or drops out, stop, reset, restart. Those break points are exactly the coordination you are building.
Silent-room simulation	30 sec	In an actually quiet room, deliver 3-4 sentences at level 1-2 as though to someone sitting next to you. Rehearses the real-world scenario, not an abstract exercise.
Volume anchoring	30 sec	Find level 2 and hold it for four consecutive sentences without drifting upward. Drift upward is the default and will happen constantly at first. Noticing the drift is the skill.

Block D — Resonance and depth (2 minutes)

Exercise	Dose	How, and what "correct" feels like
Chest hum	1 min	Hand flat on the sternum. Hum low and find the pitch where the vibration under your palm is strongest — usually the lower third of your range. That is your resonant sweet spot.
Hum-to-speech	30 sec	Start from that hum and slide directly into a sentence, trying to keep the chest buzz alive through the words. This is where perceived depth actually comes from.
Open-throat sentence	30 sec	Speak while holding the yawn-sigh posture — dropped larynx, wide pharynx, released jaw. Fingers on the throat to verify the larynx is not climbing.

Block E — Prosody and tone (2 minutes)

Exercise	Dose	How, and what "correct" feels like
Pause-for-emphasis	1 min	Take a sentence with one key word. Deliver it emphasising that word <i>only</i> by pausing before it — volume must stay completely flat. Repeat with a different key word each time.
Soft onsets	30 sec	Start vowel-initial words with a faint "h" of air ahead of the tone ("h-always", "h-I think"). Then remove the h while keeping the airy beginning. This is the main acoustic ingredient of politeness.
Emotional palette	30 sec	One sentence, two of the six modes from Section 7, rotating daily. Record and compare.

All-day background habits

- **Pause instead of amplifying.** Every time the urge to get louder for emphasis appears, insert a beat of silence instead.
- **Space between words.** Your voice needs more room around each word than most people's, because of masking. Speed costs you intelligibility at a rate it does not cost others.
- **Falling endings.** Land statements downward, and do not let sentences trail off into a clipped stop.
- **Hydration.** Consistent water intake genuinely improves fold pliability; there is no trick beyond it.
- **Larynx spot-checks.** A few times a day, two fingers on the throat mid-sentence. Riding up means the old habit has returned.

9. The weekly and monthly protocol

Cadence	What to do	Why
Daily	The 12-minute routine. Log session dB average if the app is measuring it.	Motor habits change through frequency, not through session length. Twelve minutes daily beats an hour on Sunday.
Weekly	Open MPT at habitual volume — the primary metric. Record 60 seconds of natural speech and listen back a day later.	MPT moves fast for you because it is measuring a habit, not a capacity. Listening back a day later removes the recency bias in self-judgement.
Monthly	Full retest: count, /s/, /z/, ratio, MPT loud, MPT soft. Six emotional-palette recordings back to back.	Keeps the whole picture honest and catches drift in any single dimension.
Quarterly	Re-read this document. Compare against Section 1.	The complaints list in Section 1 is the real scorecard. Numbers are proxies for it.

10. Metrics, baselines, and 12-week targets

Metric	Baseline (Aug 2026)	12-week target	Notes
Open MPT, habitual volume <i>(primary metric)</i>	15-16 sec	24-25 sec	The headline number. A proxy for effort, not for lung capacity. Should climb steadily as the pushing fades; a plateau means the habit has re-established itself.
Open MPT, soft volume	25 sec	25+ sec	Already normal. This is the ceiling the loud number is chasing. The <i>gap</i> between the two is the real measure of the habit.
Loud-to-soft gap	~10 sec	< 3 sec	Arguably the cleanest single indicator of progress in this whole document.
Session dB average	Not yet measured	-6 to -8 dB from baseline	The variable actually driving everything else. Needs the app to measure it. Establish a baseline in week 1.
Level 1 hold duration	Unstable	60 sec continuous, no dropout	The quiet-register metric.
s/z ratio	0.72	Stay < 1.4	A monitoring metric only, not a training target. If it ever crosses 1.4, see Section 13.
Count on one breath	28	Maintain	Already normal. Not a training target.
Words per minute	Not measured	Establish, then reduce ~10%	Pace matters more for you than for most people because of masking.
Emotional palette distinguishability	Anger only	All 6 identifiable blind	Subjective, but testable — play recordings to someone else.

11. What this means for the app — Speaking Lab specification

The diagnosis changes what the Speaking Lab module should actually contain. Previously it was a general delivery trainer; it now has one primary metric and one primary target.

Core principles for the build

1. **Open MPT at habitual volume is the headline number.** Show it on the home screen. It moves fast because it tracks a habit, which makes it unusually motivating compared to typical slow-moving fitness metrics.
2. **Running dB average per session is the second number.** It is the variable actually causing everything else, and it is measurable straight from the phone microphone with no extra hardware.
3. **Live feedback beats post-hoc scoring.** The single most valuable feature is a real-time volume meter that warns when you drift above your target level. Habit change requires immediate feedback, not a report at the end.
4. **Every exercise ends with a transfer rep.** Straw-then-speak, hum-then-speak, ladder-then-speak. The app should enforce this, since the transfer step is the one most likely to be skipped.

Modules to build

Module	What it does	Implementation notes
1. MPT Tracker	Guided timed test for open vowel at loud and soft volume. Charts both lines plus the gap between them.	Mic-based auto-stop when the tone ceases. Weekly reminder. The gap line is the most informative visual — make it the default view.
2. Live Volume Meter	Real-time dB display with a target band. Gentle haptic or visual nudge when you exceed the band.	Web Audio API + AnalyserNode; runs entirely client-side, no API cost. Calibrate personally in week 1 — absolute dB from a phone mic is unreliable, so use relative-to-your-own-baseline.
3. Volume Ladder Trainer	Prompts a sentence at a target level 1-5; measures whether you actually hit it; scores consistency across four sentences.	The primary fix for the quiet-register problem. Should include a "level 1 hold" timed challenge.
4. Straw / SOVT Timer	Guided 3-minute SOVT block with a mandatory transfer rep at the end.	Simple timer plus audio cue. Low effort to build; put it in phase 1.
5. Pause Trainer	Displays a sentence with one word marked. You must emphasise it using a pause only. Detects whether volume spiked on the marked word.	Detecting a volume spike is straightforward amplitude analysis — no LLM needed. High-value, low-complexity.
6. Emotional Palette	Records the same sentence in six modes; plays them back to back for comparison; stores a monthly set for longitudinal comparison.	Optionally have Gemini classify the intended emotion blind and report whether it guessed right — a genuinely good use of a free-tier LLM.
7. Pace & Masking Monitor	Words per minute plus flagging of merged word boundaries.	Speech-to-text with timestamps; flag inter-word gaps below a threshold. Ties directly to the consonant-merging complaint.
8. Larynx / Resonance Check	Records the chest hum, tracks the resonant sweet-spot pitch over time.	Pitch detection via autocorrelation. Nice-to-have; phase 3.
9. Daily Session Runner	Runs Blocks A-E end to end with timers, prompts, and a completion log.	The habit-loop shell. Should work offline — this is the module that has to be reliable.
10. Longitudinal Dashboard	MPT lines, dB average, streak, level 1 hold duration, WPM.	Keep it to five numbers. More than that stops being read.

Suggested build order

1. **Phase 1 (highest value per unit of effort):** Daily Session Runner + Straw Timer + MPT Tracker. This alone delivers most of the benefit.
2. **Phase 2:** Live Volume Meter + Volume Ladder Trainer. Directly targets the root cause and the quiet-register gap.
3. **Phase 3:** Pause Trainer + Emotional Palette. Targets the harshness and politeness problem.
4. **Phase 4:** Pace Monitor + Resonance Check + Dashboard.

Design note. Everything in phases 1–3 except the optional emotion classification runs entirely on-device with the Web Audio API. No API calls, no cost, works offline. That fits the free-tier constraint cleanly and means the core loop never depends on network availability.

12. Things that would make this worse

Do not deliberately push your pitch downward below your natural range. This is the single most common self-inflicted voice injury. It produces vocal fry, fatigues quickly, sounds forced to everyone except you, and over time can genuinely irritate the folds. Depth comes from *opening up* — never from *pressing down*.

- **Do not whisper as a substitute for speaking softly.** Whispering is more strain, not less. Level 1 must be voiced.
- **Do not use volume for emphasis** while trying to lower your average volume. The two goals are in direct conflict; pause instead.
- **Do not treat straw phonation as the goal.** Without the transfer rep it becomes a pleasant ritual that changes nothing about how you actually speak.
- **Do not judge your voice by internal feel.** Bone conduction lies to you. Record.
- **Do not chase throat-clearing.** It is a percussive impact on the folds; swallow or sip water instead.
- **Do not push through hoarseness.** If the voice is tired, semi-occluded work and rest — not more reps.

13. When to actually see a doctor

Nothing in the measurements suggests a medical problem, and this document is a training plan rather than a medical assessment. But these specific signals warrant an ENT or a speech-language pathologist rather than more self-directed practice:

- Hoarseness persisting more than two to three weeks without an obvious cause such as a cold.
- The s/z ratio crossing **1.4** on a retest — that would indicate a closure problem that was not present in August 2026.
- Pain while speaking, as distinct from effort or fatigue.
- A sudden loss of range or a voice that cuts out without warning.
- Difficulty swallowing, or a persistent sensation of a lump in the throat.
- Coughing up blood, or unexplained persistent throat pain.
- Doing the routine consistently for eight to twelve weeks with the MPT number refusing to move at all.

An SLP is also simply the fastest route even without red flags — one or two sessions of direct feedback would compress months of self-guided trial and error, particularly for the quiet-register work where the feedback loop is hardest to run alone.

14. Appendix

A. Reproducible test protocol

Run these identically each time, or the comparison to the August 2026 baseline is meaningless.

1. **Conditions:** quiet room, seated upright, at least two hours after waking, not immediately after a long stretch of talking. Hydrated. Same time of day each retest where possible.
2. **Count on one breath:** take one comfortable breath — not a maximal one — and count aloud at a normal conversational pace until you must breathe. Record the final number.
3. **Sustained /s/:** comfortable breath, then hold a steady "sssss" for as long as possible. Time it. Rest 30 seconds.
4. **Sustained /z/:** same, with "zzzzz". Time it. Rest 30 seconds.
5. **s/z ratio:** divide the /s/ seconds by the /z/ seconds. Record both raw numbers as well as the ratio — the ratio alone hides whether both values were healthy or both were short.
6. **MPT, habitual volume:** comfortable breath, then sustain an open "aaah" at your normal speaking volume, as steadily as possible, until you run out. Do not push for a record — the point is to measure the habit as it actually is.
7. **MPT, soft volume:** the same, at a volume that feels almost uncomfortably soft. Take the best of three attempts for each MPT measure.

B. Glossary

Term	Meaning
Subglottic pressure	Air pressure below the vocal folds. More pressure means more volume and more air spent. Your habitual level is too high — the root cause of everything in this document.
MPT (Maximum Phonation Time)	How long you can sustain a vowel on one breath. Here it functions as a measure of vocal <i>efficiency</i> , not of lung capacity.
s/z ratio	Sustained /s/ divided by sustained /z/. Above ~1.4 suggests the folds are not closing properly. Yours: 0.72 — comfortably healthy.
SOVT (semi-occluded vocal tract)	Any exercise that partially blocks the airway — straw, lip trill, humming, /z/. The back-pressure props the folds into efficient vibration and lowers the effort required. Demonstrably effective on you.
Pressed phonation	Voicing with excessive fold compression and pressure. Sounds harsh, spends air fast, fatigues quickly.
Masking	Loud sound energy persisting briefly and drowning out the quieter sound that follows. This is why your word-final consonants disappear and your words appear to merge.
F0 (fundamental frequency)	The actual pitch of the voice, set by vocal fold length and mass. Essentially fixed in an adult man.
Resonance	How the throat and mouth shape the sound after the folds produce it. This is where <i>perceived</i> depth lives, and unlike F0 it is trainable.
Larynx height	The vertical position of the voice box. Higher shortens the vocal tract and raises resonance (thinner). Lower lengthens it (deeper). Yours rises under load.
Bone conduction	Sound reaching your ears through your skull. It adds bass that no listener hears — which is why your voice sounds deeper to you than in a recording.
Soft onset	Beginning a word with airflow slightly ahead of the tone, rather than with a glottal click. A primary acoustic ingredient of politeness.
Prosody	The pitch, pace, pause, and rhythm layered over words. It carries emotion and attitude — and it is a trainable skill, not a fixed trait.

Compiled August 2026 from a self-administered voice assessment. This is a training reference, not a medical evaluation. Baseline numbers recorded here are the comparison point for every future retest — keep them.